



East West University

Lab Report

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Course Title: Electronic Circuits **Course Code:** CSE251

Section: 04

Expt No: 05

Expt Name: Adder Circuit using 741 Op Amp

Group No: 06

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Title: Adder Circuit using 741 Op Amp.

Objectives

1. To familiarize with the 741 Op Amp Integrated Circuit (IC).
2. To design and construct an adder using 741 Op Amp.

Theory (Summary)

Operational Amplifier (Op Amp) is a differential amplifier and can perform mathematical operations such as addition, subtraction, etc. This is an integrated circuit (IC).

The block diagram of the 741 Op Amp is shown in Figure-1. It has 8 pins. To identify the pins of the 741 Op Amp in the laboratory, place the Op-Amp on the trainer board in such a way that the notch is in your left side. Then the pin number should be identified as shown in Figure-1.

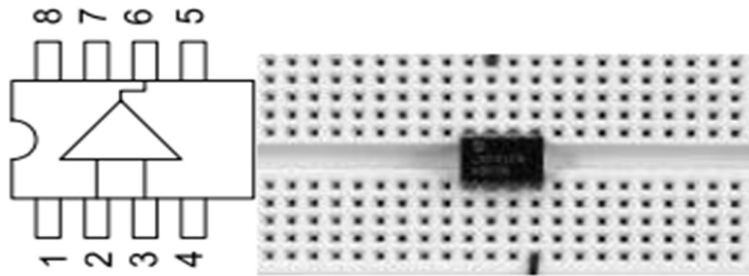


Figure-1: Block diagram of the 741 Op-Amp

The names of different pins are as follows:

- 1: Offset null (usually not used)
- 2: Inverting input terminal
- 3: Non-inverting input terminal
- 4: Negative DC power supply (usually 5-15V negative)
- 5: Offset null (usually not used)
- 6: Output terminal
- 7: Positive DC power supply (usually 5-15V positive)
- 8: Not connected (NC)

The Op Amp acts as an adder when the non-inverting input terminal is grounded and two or more signals are fed to the inverting input via resistors of appropriate values. The output becomes negative of the sum of the input signals.

A non-inverting amplifier produces the amplified output equal to the input multiplied by the closed loop gain, which is determined by the appropriate resistances.

Equipment and Component

1. Digital trainer board
2. DC power supply
3. Signal generator
4. Oscilloscope
5. Digital multimeter
6. 741 Op Amp (1 pc)
7. Resistor (as required from pre-lab design)
8. Breadboard
9. Connecting wires

Circuit Diagram

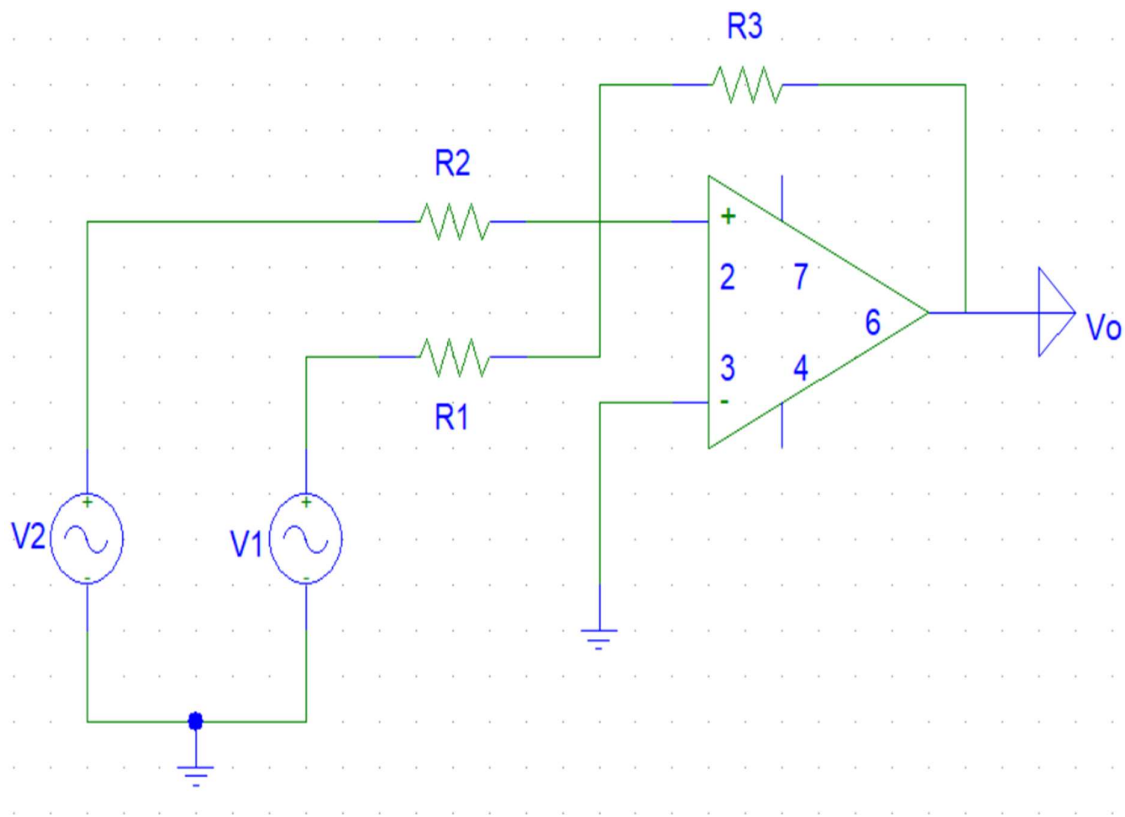


Figure-2: an adder circuits using 741 Op Amp.

Pre-Lab

1. DESIGN OF ADDER CIRCUIT: Design the adder circuit shown in Figure-2 (determine the values of the resistors) so that it implements the function $V = -(V_1 + 2V_2)$. The available set of resistances that you can use is 100 Ω , 1K Ω , 2.2K Ω , 3.3K Ω , 5.6K Ω , and 10K Ω . For design safety, take R_f as high as possible. In your design, if exact value of resistance is not available, choose the nearest one.

Answer

As we are using only 1 Op Amp, we can design the circuit as follow:

We know, For adder circuit, $V_o = -\left(\frac{R_F V_1}{R_1} + \frac{R_F V_2}{R_2} + \dots + \frac{R_F V_N}{R_N}\right)$ (i)

Comparing equation (i),

$$\frac{R_F}{R_1} = 1$$

$$\frac{R_F}{R_2} = 2$$

$$\therefore R_2 = 0.5R_F$$

Since we are required for taking R_F as high. $R_F = 10K\Omega$, $R_F = 5K\Omega$.

Since we don't have 5K Ω resistor, but we have 5.6K Ω .

$$\therefore R_2 \approx 5.6K\Omega.$$

Post-Lab

1. From the measurement in step-3, verify your design.

Answer

Here, $V_1 = 5V$; $V_2 = 2V$;

Output voltage, $V_o = -(V_1 + 2V_2)V = -(5 + 4)V = -9V$

$$V_o = -(V_1 + 2V_2)V = -(5 + 4)V = -9V$$

And from the measurement Output voltage, $V = -12.99V$

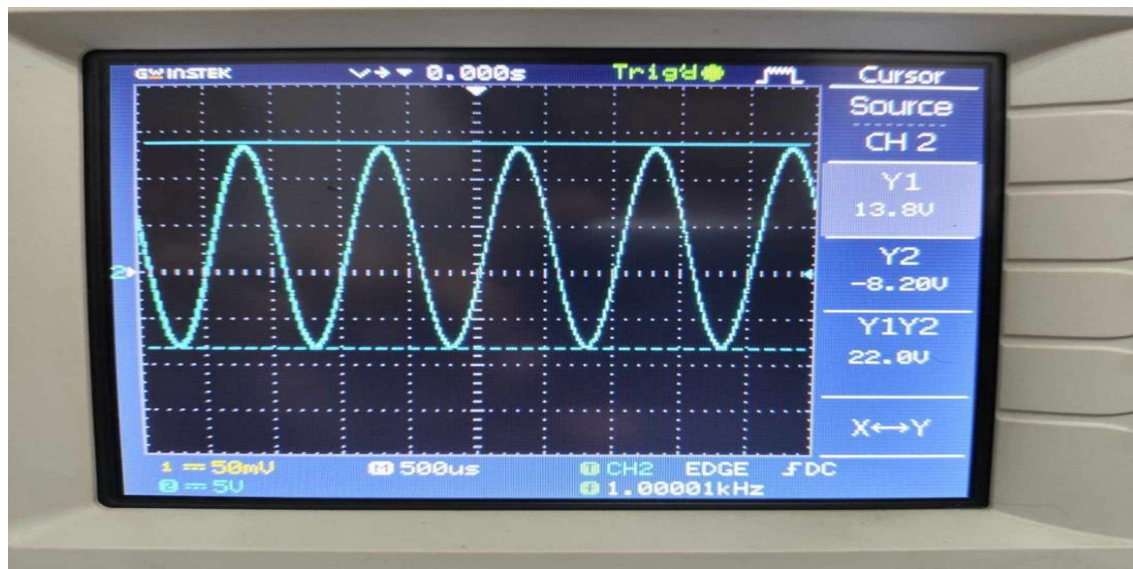
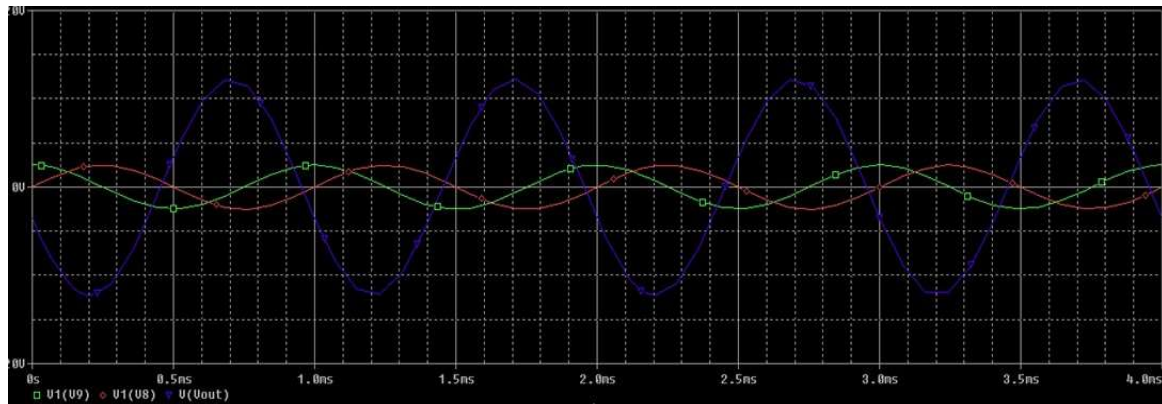
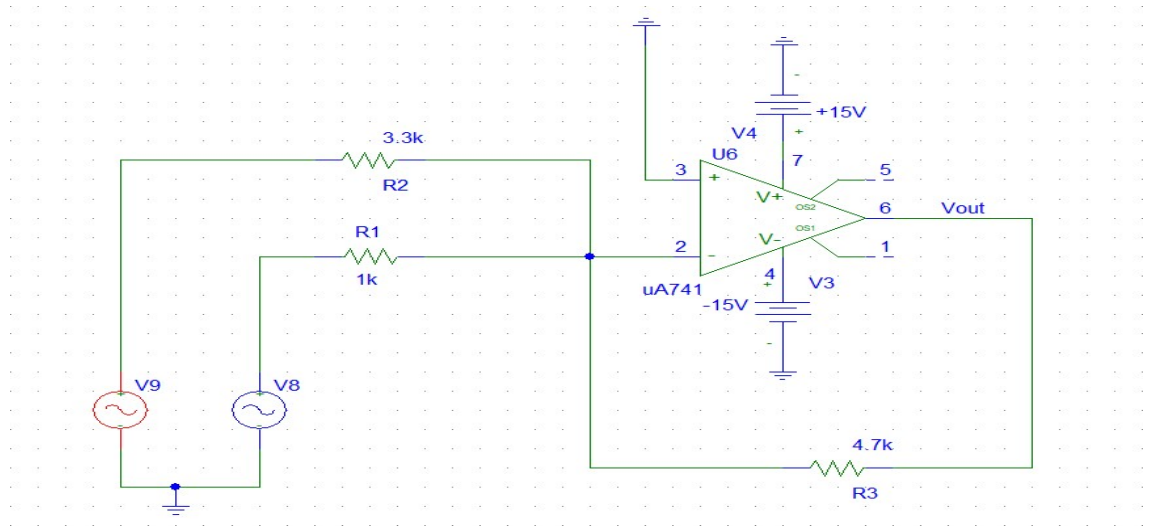
2. Does the amplitude measured in step-4 verify your design? Explain.

Answer

Amplitude measured in step 4 is 4.80V as we used 5V from AC power supply and the output taken from DC mode. So, the amplitude is almost the same here.

3. Simulate the circuit shown in Figure-2 in Pspice. Use V_1 a 5V peak to peak, 1KHZ, 0° phase sine wave and V_2 a 5V peak to peak, 1KHZ, 90° phase sine wave. Perform simulation for 4 cycles (transient analysis for 4 ms) and attach the printed output with your report.

Answer



4. Write the expressions of V_1 and V_2 of Question-3 in phasor domain, add them, and write the result in time domain as a sine function. Compare it with the Pspice output in terms of amplitude, phase angle, and time period.

Answer

In Phasor domain,

$$\text{Here, } V_1 = 2.5\angle 0^\circ \text{ V, } V_2 = 2.5\angle 90^\circ \text{ V}$$

$$V_1 + V_2 = 2.5\angle 0^\circ \text{ V} + 2.5\angle 90^\circ \text{ V} = 3.536\angle 45^\circ \text{ V}$$

So, phase angle, $\phi = 45^\circ$

Now,

$$\text{Amplitude of magnitude, } V_m = \sqrt{\{(2.5)^2 + (2.5)^2\}} = 3.536 \text{ V}$$

$$\text{Angle, } \theta = \tan^{-1}\left(\frac{2.5}{2.5}\right) = 45^\circ$$

In time domain we know,

$$V_t = V_m \cos(\omega t + \phi)$$

$$\text{Or, } V_t = V_m \cos(2\pi f t + \phi)$$

$$V = 3.536 \cos(2\pi f t + 45^\circ) \quad [f = 1\text{ kHz} = 1000\text{ Hz}]$$

$$= 3.536 \cos(2000\pi t + 45^\circ)$$

$$= 3.536 \sin(2000\pi t + 45^\circ + 90^\circ)$$

$$= -3.536 \sin(2000\pi t + 135^\circ)$$

We know,

$$\text{Time period, } T = \frac{1}{f} = \frac{1}{1000} = 10^{-3} \text{ s} = 1\text{ ms}$$

From PSpice simulation,

We find amplitude = 4.80V, Phase angle = 90° And time period = 1ms

Comparison between PSpice and theoretical calculation:

	From PSpice	From calculation
Amplitude	4.8V	4.8V
Phase angle	90°	45°
Time period	1ms	1ms

Discussion

The purpose of this experiment was to design and build an adder circuit using a 741 Op Amp and to become familiar with the 741 Op Amp integrated circuit that can perform mathematical operations. In this experiment, the output is the negative sum of the input signals.

The experiment results demonstrated a discrepancy between the theoretical calculations and the measured values for output voltage V_o . This difference may be due to component tolerances, power supply fluctuations, measurement errors and the non ideal behavior of the 741 Op Amp.

In Pspice results for sinusoidal inputs, the theoretical time period aligns perfectly with the simulation result. However, there is a significant difference for both the amplitude and the phase angle (calculated 45° vs. Pspice 90°) which suggests a potential issue in the simulation setup or could be due to the misunderstanding of how the Pspice circuit was implemented or a potential error in the calculation.

Despite these discrepancies, the experiment successfully demonstrated the fundamental principals of using an Op Amp as an adder. This laid to a foundation for more advanced circuit design.